

MUHAMMAD TAHA BIN ZAEEM

Computer Engineering Undergraduate | AI Builder | Student Community Organizer

mtaha.ce47ceme@student.nust.edu.pk | tahabinzaeem.tech | GitHub | LinkedIn | Pakistan

PROFILE

Computer Engineering undergraduate at NUST CEME combining **AI building** with **student community leadership**. Founder of Type2Learn and ProGenEDA, built through intensive AI-assisted development workflows and then strengthened with structured engineering and validation. **Volunteered in organizing 5+ technical workshops in my first year** and now work with our Head of Department to establish a recurring workshop culture for new incoming batches, with **at least 10 workshops planned for Winter Semester 2026**. Focused on helping students move from AI curiosity to practical, responsible building.

CAMPUS COMMUNITY & WORKSHOP LEADERSHIP

Technical Workshop Organizer & Department Initiative — NUST CEME

2025 - Present

- Volunteered in organizing 5+ technical workshops during my first year, building hands-on experience in student-led technical programming.
- Currently collaborating with the Head of Department to establish a repeatable culture of technical workshops for incoming batches and future academic years.
- Working toward a first concentrated run of at least 10 workshops during Winter Semester 2026.

SELECTED AI BUILDER PROJECTS

Type2Learn — Accessibility-First AI Learning Platform

Founder & Development Lead | Jul 2026 - Present | AI-native / vibe-coded build

- Built a bilingual active-learning platform with learner-controlled accessibility supports and course-aware AI explanations, hints, simpler framings, and examples.
- Finalist, P@SHA ICT Awards 2026 — currently competing for Gold at Pakistan's premier ICT awards.

ProGenEDA — Hybrid AI + Deterministic EDA Generation

Founder, Product & Engineering Lead | Jul 2026 - Present | AI-native / vibe-coded build

- Turns bounded circuit intent into structured CircuitIR/JSON, then uses deterministic validators and exporters to generate editable Proteus, KiCad, EasyEDA Pro, and LTspice projects.

ParetoCo — AI-Assisted Computer-Architecture DSE

Co-creator | 2026 - Present

- Co-built a research environment combining native constraint search, workload modeling, multi-objective analysis, and bounded LLM assistance for architecture exploration.

TECHNICAL CREDIBILITY

Custom 20-bit Single-Cycle Vector CPU — NUST Academic Team Project

- Co-designed a Harvard-architecture processor with 64-bit SIMD operations across four 16-bit lanes; project earned Second Runner-Up in the SEMPEC Junior Hardware category.

Applied AI Research

- Current work includes controlled LLM rewriting evaluation, reproducible data pipelines, and AI-assisted engineering workflows where deterministic computation remains the source of truth.

EDUCATION

NUST, CEME

B.E. Computer Engineering

Aug 2025 - Oct 2029 (expected)

Third semester

LEADERSHIP

Vice President

Dr. Samar Mubarakmand Society for Sciences, GCU Lahore

Sep 2023 - Jan 2025

Vice President

Dr. Khorana Society for Chemistry, GCU Lahore

2023 - 2025

HONOURS

- P@SHA ICT Awards 2026 Finalist — Type2Learn
- Second Runner-Up, SEMPEC Junior Hardware — Vector CPU
- Runner-Up, STEM 2024 — PKR 20,000 prize

EXPERIENCE

PhishRod

Development Intern

Jul 2026 - Present

Arch Technologies

Software Programmer Intern

Jan 2026 - Apr 2026

SKILLS

AI / ML: Python, NumPy, Pandas, Jupyter, LLM prompting, structured outputs, evaluation

Build: FastAPI, React, JavaScript, Git, Linux

Engineering: C++, Verilog, MIPS, Proteus, KiCad, LTspice, Xilinx ISE

Research: Dataset design, reproducible pipelines, experimental controls, technical writing

SELECTED CREDENTIALS

- Machine Learning Specialization — DeepLearning.AI / Stanford Online
- Logic & Critical Thinking Specialization — Duke University
- Game Theory — Stanford / UBC

CAMPUS PROGRAM GOAL

Build a durable Claude community at NUST through practical AI workshops, prompting sessions, no-code buildathons, responsible experimentation, and student-led building.